Example 2.7

Find z^{-1}, if $z=(2+3i)(1-i)$.

Solution

We have $z=(2+3i)(1-i)=(2+3)+(3-2)i=5+i$

$$\Rightarrow \quad z^{-1}=\frac{1}{z}=\frac{1}{5+i}\ .$$

Multiplying the numerator and denominator by the conjugate of the denominator, we get

$$z^{-1}=\frac{(5-i)}{(5+i)(5-i)}=\frac{5-i}{5^2+1^2}=\frac{5}{26}-i\frac{1}{26}$$

$$\Rightarrow \quad z^{-1}=\frac{5}{26}-i\frac{1}{26}.$$

Example 2.8

Show that (i) $\left(2+i\sqrt{3}\right)^{10}+\left(2-i\sqrt{3}\right)^{10}$ is real and (ii) $\left(\frac{19+9i}{5-3i}\right)^{15}-\left(\frac{8+i}{1+2i}\right)^{15}$ is purely imaginary.

Solution

(i) Let $z=\left(2+i\sqrt{3}\right)^{10}+\left(2-i\sqrt{3}\right)^{10}$. Then, we get

$$\overline{z}=\overline{\left(2+i\sqrt{3}\right)^{10}+\left(2-i\sqrt{3}\right)^{10}}$$

$$=\overline{\left(2+i\sqrt{3}\right)^{10}}+\overline{\left(2-i\sqrt{3}\right)^{10}} \qquad \left(\because \overline{z_1+z_2}=\overline{z_1}+\overline{z_2}\right)$$

$$=\left(\overline{2+i\sqrt{3}}\right)^{10}+\left(\overline{2-i\sqrt{3}}\right)^{10} \qquad \left(\because \overline{(z^n)}=(\overline{z})^n\right)$$

$$=\left(2-i\sqrt{3}\right)^{10}+\left(2+i\sqrt{3}\right)^{10}=z$$

$$\overline{z}=z \Rightarrow z \text{ is real.}$$

(ii) Let $z=\left(\frac{19+9i}{5-3i}\right)^{15}-\left(\frac{8+i}{1+2i}\right)^{15}$.

Here, $\frac{19+9i}{5-3i}=\frac{(19+9i)(5+3i)}{(5-3i)(5+3i)}$

$$=\frac{(95-27)+i(45+57)}{5^2+3^2}=\frac{68+102i}{34}$$

$$=2+3i\ . \qquad (1)$$

and $\frac{8+i}{1+2i}=\frac{(8+i)(1-2i)}{(1+2i)(1-2i)}$

$$=\frac{(8+2)+i(1-16)}{1^2+2^2}=\frac{10-15i}{5}$$

$$=2-3i\ . \qquad (2)$$

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